

Fig. 6: Author's prototype

For testing the circuit, proceed as follows:

1. After assembling all the components on the PCB, connect TX and RX pins of the GSM modem to pins 13 and 14 of MAX232, respectively. Insert a valid SIM in the card holder of the GSM modem.

2. Connect ground pin of the GSM modem to the ground rail of the circuit.

3. Use two single-strand (hook-up) wires as sensor A and sensor B. (In the

actual application, use of steel rods as sensors is recommended.) Hang the sensors into a bucket or mug such that sensor A is above sensor B.

4. Pour water into the bucket until the water level reaches sensor A.

5. Now switch on the circuit. You should get a call on your mobile phone. This indicates that you can turn the motor 'on.'

6. Send SMS "motor11 on" from any mobile phone to the SIM in the modem to turn the motor 'on.' You

can also turn the motor 'off' by sending "motor11 off" message from any mobile phone.

7. Now remove water from the bucket until the water level in the bucket dips below sensor A. The motor should automatically switch off and you should receive a call from the modem simultaneously alerting you that the borewell (bucket in this case) is empty and the motor has been switched off.

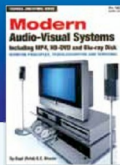
After testing the above steps, you can install the system in the borewell by inserting sensors A and B into the pipe with sensor B placed at the bottom of the borewell as shown in the circuit. Your borewell monitoring system is now ready for use. The author's prototype is shown in Fig. 6.

EFY note. The source code of this article is available on www.efymag.com.

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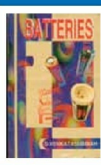
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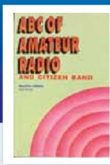
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